Chained transfer differential reference

2026-11-11 · fixed F_{101} fixture · producer-run development checks

Standing: unverified. The reference arithmetic predates this work and is imported unchanged, but both implementations are run by this producer and use Cantor reduction. Agreement is not signed independent verification.

Pinned source and models

Native source: isogeny-labs/crypto commit 18db046d2e9fd0b01473f549815a1b51069da677, package research/chained_jacobian_transfers_20261110/native. The reference is suite/src/prime_hyperelliptic from cryptanalysis commit 68594c8874c75a0a6c04a3874adb1374088ca4f5. source-pin.json and SHA-256 manifests bind all imported files; check.sh refuses drift before compiling.

The source is $C: y^2 = x^6 + 1$ over F_{101}, based at (10,0). The target is $C_2: w^2 = 41z^6 + 9z^4 + 9z^2 + 41$. The order-17 elliptic subgroup has P=(4,41) on $E: y^2 = x^3 + 1$. A faithful coordinate change makes each odd quintic monic: $t = a \cdot z$, $Y = a^2 \cdot Z$ with a=inverse(leading(F)). The reference therefore computes on the same curves, not an unrecorded twist. [pcc:unverified]

Exact development coverage

Check	Retained scope
Source arithmetic	10,404 canonical additions and inverses
Richelot subgroup	All 289 additions in the order-17 target subgroup
Full forward image	10,404 source images and fixed-translation additions; 2,601 distinct target classes
Point-count order	Both Jacobian orders recomputed as 10,404 from F_p/F_{p^2} counts
Local suite	26 tests pass; 1 pre-existing benchmark remains ignored

Raw outputs are evidence/differential-01.txt through differential-04.txt and the pinned-run receipt. First Clippy failure and the subsequent pass are retained; its compatibility annotation affects only the unchanged reference's from_fp(&self) API. No correctness test was suppressed. [pcc:unverified]

Reproduce and limits

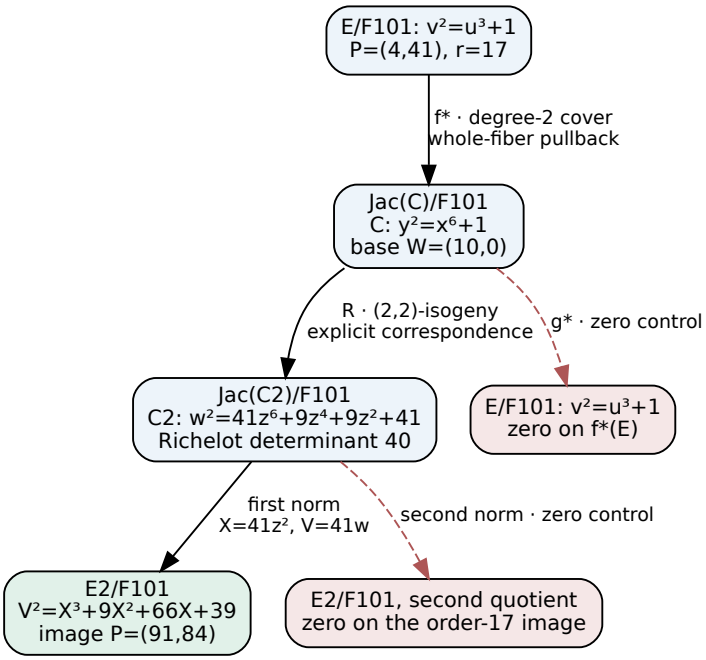
Check out the two repositories as siblings, with the pinned native files in crypto, then run experiments/chained-transfer-reference/check.sh from cryptanalysis. Cargo.lock and numeric dependency versions are fixed.

There is no DLP timing or speed claim, new curve-family coverage, or full formal Picard construction in this record. The upstream report retains vanishing projection controls and the elliptic shortcut comparison. These checks do not turn that route into a new computational reduction. [pcc:unverified]

Primary sources: Galbraith, corrected book ch.8 §8.3 and ch.10 §§10.3–10.4; Kunzweiler, DCC92 (2024), DOI10.1007/s10623-024-01366-1, §§3.1–3.2. The complete source receipt is pinned upstream. KB and vault gaps remain explicit; no novelty claim is made.

Route of the pinned upstream fixture

F101 chained Jacobian fixture · producer development evidence · PCC unverified



This diagram is copied from the pinned upstream commit. Solid arrows show the explicit maps; dashed arrows are zero controls. All evidence remains unsigned.